



Computer vision-based non-contact structural vibration measurement: Methods, challenges and opportunities

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ABSTRACT

Understanding and optimizing system performance hinges on analyzing structural vibrations. While contact-based vibration measurement faces limitations including load interference, restricted frequency response, and complex setup non-contact methods utilizing computer vision offer a superior alternative. These methods boast high accuracy, comprehensive measurement capabilities, and cost-effectiveness. This work reviews the latest research in computer vision-based non-contact vibration measurement, exploring both its potential and its challenges. We begin by outlining the fundamental principles and common optical measurement methods. This is followed by an examination of point tracking and digital image correlation, including their technical foundations and applications. Finally, using real-world examples, we discuss current challenges and propose innovative solutions to advance the field of optical vibration measurement.

1. Introduction

Structural vibrations are a commonly observed phenomenon in daily life, such as the vibration of bridge girders [1], bridge bearings [2], offshore wind turbines [3], rotating machines [4], and engines [5], etc. However, not all vibrations are beneficial. For example, the collapse of the Tacoma Narrows Bridge resulted from structural vibration resonance subject to high wind speeds. Moreover, intense vehicle vibrations can reduce driving comfort [6], causing discomfort to passengers and even injuries. Engineering structures exposed to prolonged severe vibrations are susceptible to fatigue damage, thereby shortening their lifespan [7]. Therefore, effective measurement and monitoring of the vibration dynamics are crucial in understanding the physical characteristics of engineering structures and also help validate and modify analytical models [8].

According to different techniques, principles, and installation methods used to measure the vibration of a target object, the vibration measurement methods can be categorized into two groups: the contact and non-contact methods [9]. The contact vibration measurement is mature and widely applied in various applications [10,11]. This method

primarily involves attaching accelerometers to monitor the object's vibrations. The types of accelerometers include piezoelectric, piezoresistive, and capacitive sensors [12]. The piezoelectric sensors utilize the piezoelectric crystals or ceramics as sensitive elements, generating voltage signals in response to engineering vibrations [13]. These sensors excel in high-frequency response but exhibit poor performance in low-frequency response (e.g., below 1 Hz) [14]. The capacitive accelerometer measures the voltage changes across the capacitor electrodes to produce better performance in low-frequency response and presents good ability to reduce the sensor cost, size, and power consumption [15,16]. However, the capacitive sensors may experience zero-offset drift and are sensitive to environmental interference, which requests frequent calibration and complex tuning processes [17,18]. The piezoresistive accelerometers are attractive because of their simple construction, easy manufacturing, and straightforward circuitry for readout. They offer superior full-range performance and electromagnetic interference resistance compared to the capacitive accelerometers [19]. However, they are typically subject to lower sensitivity [20], mass-loading effects in lightweight structures [21,22], and higher installation costs in large structures [23]. In summary, each type of contact sensor

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has specific advantages and limitations, and the choice of the appropriate sensor depends on the specific application requirements and environmental conditions.

Non-contact measurement is primarily divided into electromagnetic wave measurement, acoustic wave measurement, and optical measurement [24]. Electromagnetic wave measurement involves transmitting fundamental or multi-frequency electromagnetic signals and analyzing the spatiotemporal characteristics of the received signals to extract vibration information from target objects. This method often utilizes radar systems such as microwave radar [25] and millimeter-wave radar [26], enabling global measurement but susceptible to interference from multiple vibrating objects [27]. The acoustic wave measurement encompasses two forms; one uses microphone-type sensors to receive vibration acoustic signals, which are susceptible to environmental sound field interference [28] and require sophisticated signal processing [29]; the other uses ultrasonic waves [30] to offer good penetration capability but low accuracy and strict environmental requirements [31]. The optical measurement is divided into laser measurement and computer vision measurement. The former often uses laser displacement sensors and Doppler vibrometers. The laser displacement sensors are used for low-frequency vibration measurement [32,33], while the Doppler laser vibrometers are preferred for high-frequency vibration [34,35]. However, these mentioned methods are typically suitable for single-point measurements and require expensive specialized instruments.

With the rapid development of computer vision technology, optical non-contact vibration measurement is increasingly utilized thanks to its superior accuracy, easy setup, cost-effectiveness [36,37], and non-load impacting advantages [38,39]. This review provides a comprehensive survey and summary of the algorithmic principles and applications of computer vision-based non-contact vibration measurement. The main contributions of this review are described as follows:

- (1) This article classifies visual vibration measurement methods into artificial marker and natural feature methods, as well as point tracking and digital image correlation methods, based on human involvement and their underlying principles. It also examines the interrelationships of these methods in practical applications.
- (2) This article further focuses on point tracking and digital image correlation methods, providing a detailed overview of their derived techniques. It analyzes the characteristics of each method based on different application scenarios and performance metrics, offering valuable insights for practical applications.

- (3) This article summarizes the main challenges faced by future research directions based on the strengths and weaknesses of existing methods and provides corresponding prospects for development.

2. Basic Concepts of optical vibration measurement

Optical vibration displacement measurement employs imaging sensors like charge-coupled devices (CCD), complementary metal-oxide-semiconductor (CMOS) sensors, and infrared cameras to capture sequences of images of the objects. These images are subsequently processed to identify the point coordinates, patterns, features, or object boundaries, thereby enabling the calculation of vibration displacement [8]. The optical vibration displacement measurement system comprises essential hardware and software components. The hardware typically includes one or more cameras, zoom lenses adjusted for optimal magnification or reduction based on the measurement distance, and a computer interface. The software components encompass sophisticated algorithms encapsulated for real-time image processing, dynamic display of vibration displacement, and facilitating automatic data storage [40]. As depicted in Fig. 1, in an optical vibration measurement the optical camera extracts the structural profile of a bridge by measuring displacement at multiple points. The optical camera interfaces with a computer equipped with dedicated image acquisition and analysis software, supporting comprehensive analysis of the target object's vibration dynamics. Optical vibration measurement is the process of extracting and tracking real-time positional information of the target points. The strategic placement and selection of appropriate reference points are crucial for improving the accuracy of target point acquisition, thereby enhancing the precision of the vibration measurement.

2.1. Feature point detection

The point-extracting process mainly adopts two methods: the artificial marker method and the natural feature method.

- (1) **The artificial marker method**, also known as the targeted approach, involves installing high-contrast optical targets such as circular dots or image speckles on the test objects. The cameras capture the image sequences of these markers to obtain their coordinate data. Image processing algorithms are then employed to precisely determine the centers of these optical markers,

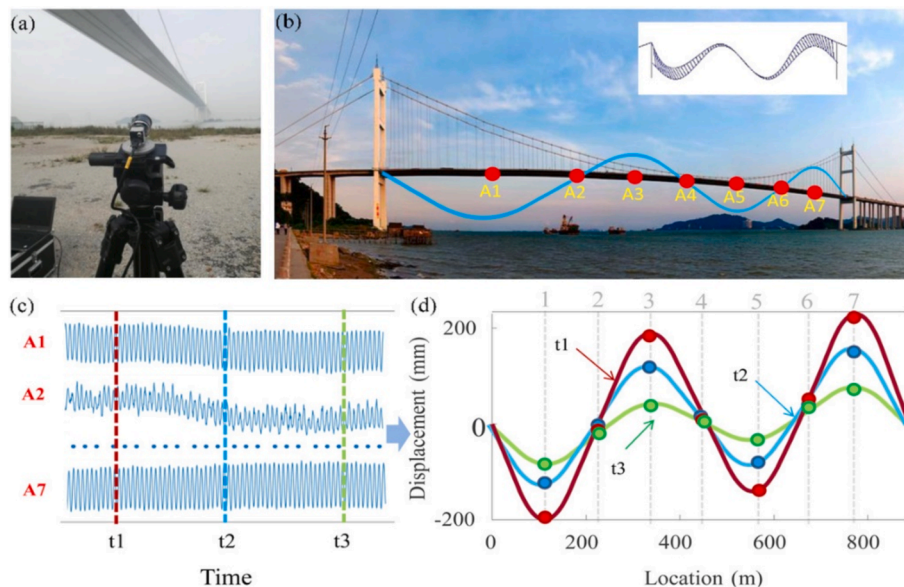


Fig. 1. Schematic diagram of typical optical vibration measurement [41].

providing the positional information for subsequent analysis [42].

- (2) **The natural feature method**, also referred to as the untargeted approach, utilizes the inherent surface features of the object itself as regions of interest. This method avoids the need for artificial markers, thereby reducing labour costs and improving measurement efficiency. Image processing algorithms detect and track these natural features based on their distinctive characteristics to calculate the vibration characteristics.

2.2. Target tracking

Both artificial marker-based and natural feature-based methods initially rely on the selection and positioning of the target points or regions. In this process, the relevant optical algorithms play a crucial role, categorized into the point tracking method and the digital image correlation (DIC).

- (1) **The point tracking method** focuses on tracking specific points of interest (such as feature points or markers) to estimate the motion of the target by detecting and tracking the positions of these points over time, which is independent of grayscale information within the region. This method typically analyzes individual or a small number of feature points and can accurately track their motion trajectories and changes.
- (2) **The DIC** calculates the deformation and displacement of an object's surface using the brightness or grayscale patterns of pixels in the image. This method relies on a well-defined DIC window and measures the displacement fields and deformation by comparing the grayscale variations of the DIC window between different image frames. The DIC usually handles large regions by enabling full-field measurements and obtaining deformation maps of the object's surface.

These mentioned optical techniques leverage advanced image processing and computer vision technologies to play a significant role in engineering applications, particularly in scenarios requiring high-precision measurements, such as structural health monitoring in civil engineering and vibration analysis in mechanical engineering. More specifics will be detailed in Section 3.

3. Methods for optical vibration measurement

3.1. Artificial marker methods

To improve the robustness of target tracking and reduce measurement errors, it is common to affix high-contrast artificial targets on the surface of engineering structures [43]. The artificial marker method is primarily divided into passive optical markers and active optical markers. The passive optical markers involve objects coated with reflective materials, such as circular targets [44], geometric panels with dots [45,46], cross shapes [47], checkerboards [48], random black-and-white blocks [49], and speckle patterns [50]. Using the image processing algorithms (e.g., the point tracking and DIC methods), the positional information of these optical target points can be accurately extracted, tracked, and recorded. For example, Wu et al. [51] installed high-contrast circular markers on a laboratory structure for vibration measurements. They discussed the maximum density limit of the markers and the relationship between the image sensor size and the field of view. Chen et al. [52] uniformly applied 254 circular *retro*-reflective targets with a 20 mm diameter on the surface of the solar panel arrays to validate the dynamic models of the solar cells using the vibration measurement data. Ji et al. [53] utilized the black-and-white grids as markers to develop an optical-based method for comprehensive deformation and crack measurement.

Due to challenges with traditional passive optical markers under

nighttime or complex lighting conditions, many researchers have turned to active optical markers. Examples include LED sources [54], laser point sources [55], and laser speckle patterns [56] (as shown in Fig. 2). For instance, Wahbeh et al. [57] installed LED markers on the bridge structures and used the optical measurement system to directly measure the displacement of the selected locations. Dong et al. [58] installed LED markers on the engineering structures to measure the dynamic response and identify the modal parameters. To be cited but too many, the artificial marker method is still a mainstream tool for optical vibration measurement.

3.2. Natural feature methods

The natural feature method, also known as a non-targeted method, detects vibrations and deformations by identifying internal characteristics or edges of the objects, without relying on optical targets or patterns. Unlike the artificial marker, it proves more convenient when optical targets cannot be installed on the object's surface or graphical optimization is impractical [59]. Similar to the marker-based method, point tracking or DIC are required for the surface feature extraction, as shown in Fig. 3. In point tracking, high-contrast structural features must be identified even without specific optical markers. The DIC requires rich-textured natural patterns on the object's surface without the need for optical speckles [60].

The non-targeted methods generally have lower computational complexity compared to the marker-based methods. However, their notable drawback lies in inferior accuracy and robustness relative to the targeted methods; particularly when the image clarity is poor or there is low contrast between the background and objects, affecting the tracking of internal features. To eliminate the reliance on artificial optical targets, many researchers have endeavoured to successfully track object surface displacements without installing artificial targets [61–66]. For instance, Lee et al. [67] chose the bridge surfaces as the natural feature region and utilized the laser radar and cameras to measure the structural displacement. Khuc and Catbas [68] used the bolt connections in the stadiums as the natural feature points for vibration measurement. Currently, the natural feature method is another reliable tool for optical vibration measurement.

3.3. Point tracking methods

Point tracking is a computer vision and image processing technique used to track specific target points or feature points across consecutive frames of images or video, to obtain spatial and temporal motion information of the target. It is particularly significant in applications such as vibration analysis and motion tracking. The process involves the following steps.

- Step 1. Feature point detection: Initially, it is necessary to detect points of interest in the initial frame.
- Step 2. Feature point description: Each detected feature point generates a descriptor that encapsulates local feature information around that point.
- Step 3. Target tracking: Once the feature points are successfully described across consecutive frames, the algorithm can track their motion trajectories, enabling target tracking and measurement.

Since the description of feature points relies on subsequent computations, the details of feature point detection and object tracking are introduced.

3.3.1. Feature point detection

In computer vision-based vibration measurement, the feature points are crucial for analyzing the dynamic response of the objects. These points typically include the corners, edges, and areas with significant grayscale changes that can be stably tracked across consecutive video

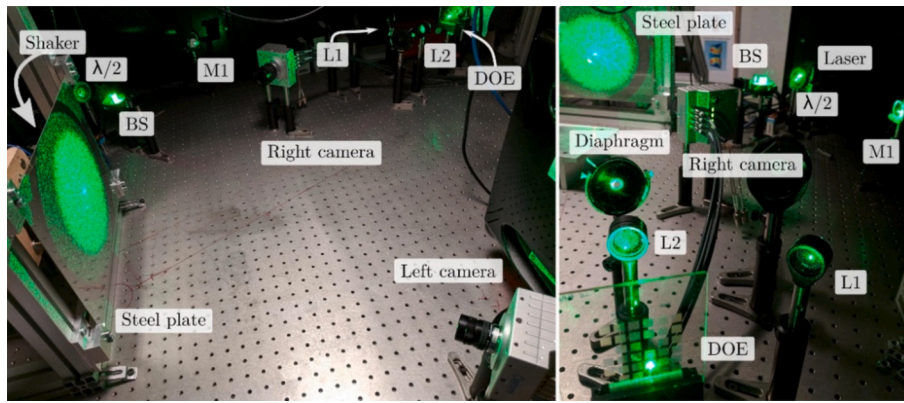


Fig. 2. Laser speckle [56].

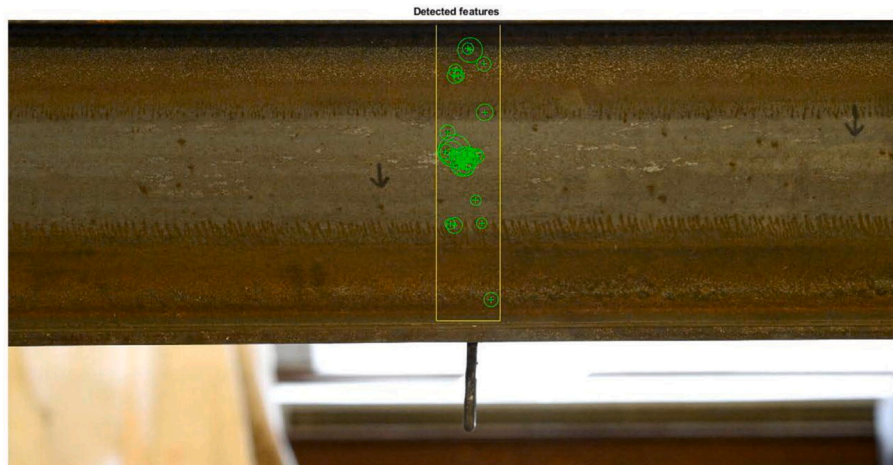


Fig. 3. Features detected in laboratory trials [65].

frames. They are highly efficient in localization and tracking motion trajectories and are commonly used for monitoring equipment status and structural vibration characteristics. The feature point detection algorithms can be divided into three main categories: edge points, gray-scale intensity, and fully data-driven methods.

Edge points-based detection is a core technology in the field of image processing, primarily used to locate and describe the contours or structures of objects by identifying areas in an image where there is a sharp change in gray levels. These areas, known as edges, are typically places where the pixel gray values change rapidly. The edge point detection further refines this to the specific pixels that make up the edges, which is crucial for accurately defining the edge's precise location and shape. The edge detection algorithms can generally be categorized into the following types.

- (a) **Gradient-based methods:** These methods detect the edges by calculating the first derivative of the image to find where the brightness changes most significantly. Common gradient-based edge detection algorithms include the Sobel operator, Prewitt operator, and Roberts operator [46]. The Sobel operator utilizes two 3x3 matrices to approximate the gradient in the horizontal and vertical directions; the Prewitt operator is similar to the Sobel operator but with different coefficients in the matrix, resulting in subtle differences in edge response; the Roberts operator uses a 2x2 matrix to calculate the gradient along diagonal directions, which is sensitive to fine edges. Among existing edge detection algorithms, the Sobel operator is widely used in vibration measurement [46].

- (b) **Second-derivative methods:** These methods detect the edges by locating points in the image where the second derivative goes to zero, precisely pinpointing the center of the edges. The Laplacian operator, as a representative of the second-derivative method, directly calculates the image's second derivative but is sensitive to noise. To solve this issue, the Laplacian of Gaussian operator (LOG) combines the Gaussian smoothing and the Laplacian operator to effectively enhance the accuracy and noise resistance of the edge detection. For example, the LOG operator has been used for edge feature extraction in rotating machinery fault diagnosis [69].
- (c) **Canny method:** This algorithm precisely identifies and locates the image edges by integrating multiple key steps, including image smoothing, gradient calculation, non-maximum suppression, dual thresholding, and edge linking [70,71]. The Canny method not only effectively suppresses the noise but also ensures the clarity and continuity of the edges. Thus, it is extensively applied in the field of optical vibration measurement. Many studies have used the Canny method to precisely extract critical structural edge information to further calculate the vibration characteristics [52,72].

With the rapid advancement of machine learning technology, edge detection methods have progressively evolved to incorporate deep learning techniques. Compared to traditional algorithms like the Sobel or Canny operators, deep learning methods are more effective in handling complex scenarios and noise issues, providing more precise and robust edge detection results. This progress has led to the

widespread application of deep learning in the field of optical vibration measurement. For instance, Wang et al. [73] introduced a novel edge feature extraction method for the bridge cables, as shown in Fig. 4. The Resnet-34 and Swin networks were used as the encoders for feature extraction, combined with a cross-model grafting module for decoding. The extract edge feature information was further used for vibration characteristic analysis. Song et al. [74] described a new optical vibration measurement method for laboratory-supported beams. This method first isolated the beam from the background noise using a fully convolutional network (FCN) and a conditional random field (CRF); subsequently, the displacement measurement was conducted using the DIC. These studies demonstrate that deep learning not only enhances the accuracy and robustness of edge detection but also expands its application scope in engineering and scientific research domains.

In the grayscale intensity, corner detection [75] utilizes information about changes in pixel grayscale intensity within an image to identify the corners. In the fields of digital image processing and computer vision, the corners refer to points within localized areas of an image where the grayscale values exhibit significant variations, often manifesting as intersections of edges or points of directional change within a region. These corners are prominent features in images, displaying pronounced changes in multiple directions. Corner detection is a fundamental task in computer vision as these points serve as the pivotal local features in images, which is essential for applications such as image matching, object tracking, and scene recognition.

As for vibration measurement, corner detection techniques such as the SUSAN (small univalue segment assimilating nucleus) and Harris corner detectors are widely employed. These methods utilize the changes in the grayscale intensity within images to identify key feature points. For instance, Zhou et al. [75] proposed an improved SUSAN corner detector to rapidly and accurately identify critical features of large gear structures for six-degree-of-freedom pose estimation. Yoon et al. [76] utilized the Harris corner detection method to extract the natural feature points from the surfaces of engineering structures and tracked these points in real-time to calculate the vibration information. Similarly, Tan et al. [77] and Tang et al. [78] used Harris corner

detection to observe the natural features of engineering structures and measure their vibration displacement. Despite potential errors in the Harris corner detection due to changes in distance, these issues have been addressed with the introduction of the SIFT (scale-invariant feature transform) method [79]. The SIFT can locate key points across different scale-spaces and provide more stable feature descriptions. Furthermore, a streamlined version of the SURF (speeded-up robust features) algorithm has found widespread application in vibration testing [65,80]. The SURF corner detection can identify and track the natural features of architectural structures and distant mechanical systems for vibration analysis, leading to the development of inexpensive non-contact vibration measurement systems.

Additionally, as an improved version of the FAST (features from accelerated segment test) algorithm, the ORB (oriented FAST and rotated binary robust independent elementary features) algorithm, offers rotational and scale invariance, making it suitable for scenarios requiring high real-time performance. With the advancement of deep learning, there have been developments in key-point detection, such as the SuperPoint algorithm [81]. Combining the key-point detection and matching algorithms [82,83], a computer vision-based method for full-field three-dimensional (3D) vibration displacement measurement was proposed, as shown in Fig. 5.

According to the open literature, the corner detection techniques applied in vibration measurement not only provide the capability to identify and track critical features but also offer reliable tools for precise vibration analysis.

Fully data-driven (Deep learning) methods: The aforementioned optical vibration displacement measurements are typically conducted in ideal optical environments to avoid the detrimental effects of environmental uncertainties on measurement accuracy and stability. Furthermore, most applications and experiments have relatively short measurement durations, with limited research on optical variations in natural environments [71], resulting in poor adaptability. In practical outdoor applications, optical vibration measurement methods are highly susceptible to optical environmental influences such as changes in lighting, shadows, atmospheric disturbances, and gusts of wind [84].

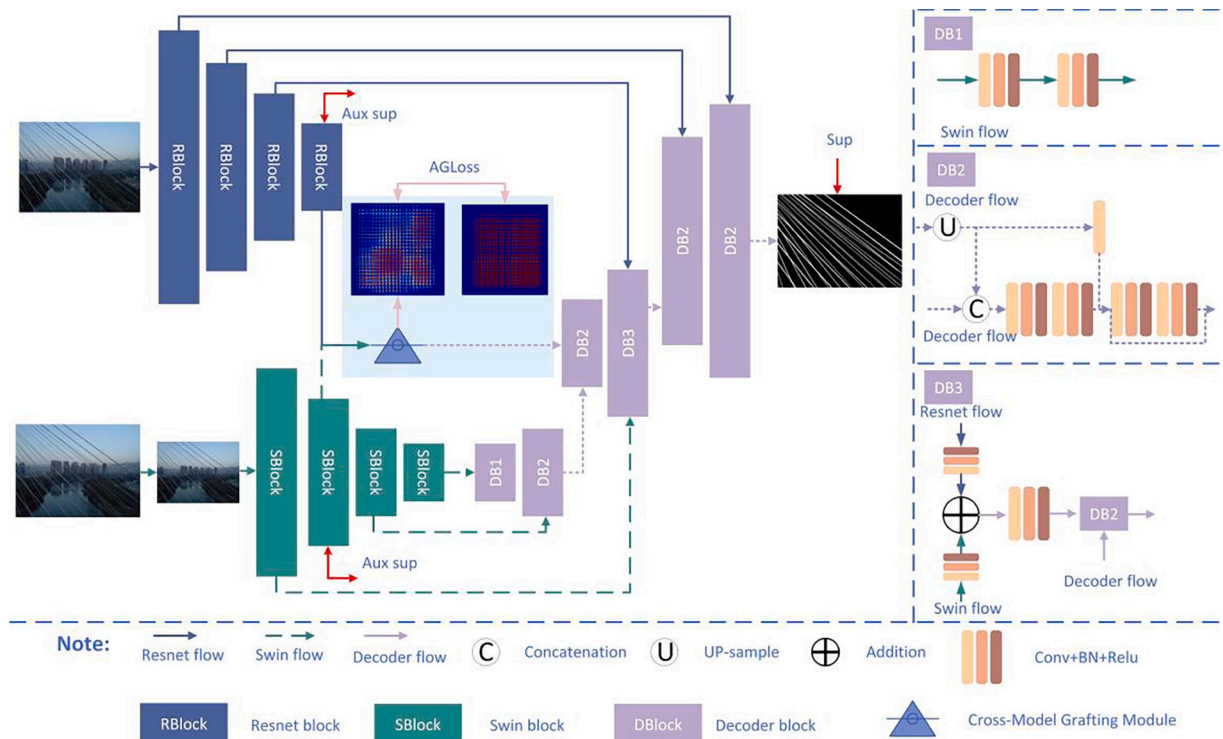


Fig. 4. Network structure of the cable background segmentation (CBS) model [73].

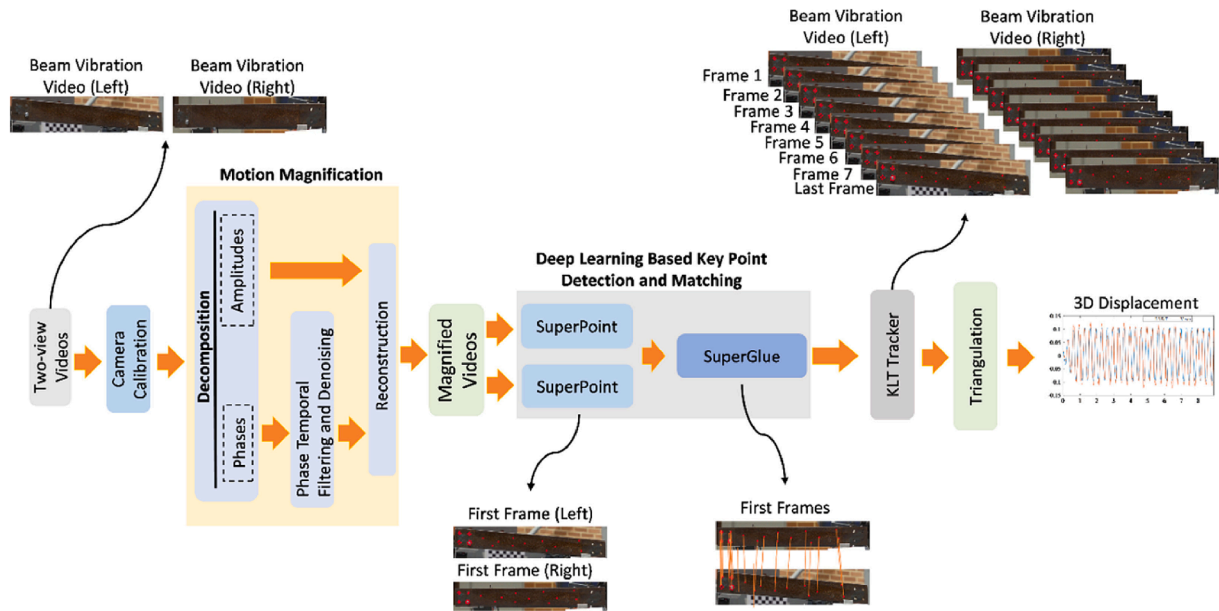


Fig. 5. 3D tiny displacement measurement method [83].

With the advancement of deep learning, convolutional neural networks (CNNs) have been continuously improved and gradually applied to enhance measurement robustness in the field of vibration measurement.

In recent years, CNNs have found widespread application in engineering problems, such as image classification, object detection, and segmentation [85,86]. Algorithms include the single-stage YOLO (you only look once) series [87–89] and SSD (single shot multibox detector) series [90], as well as the R-CNN (region-based CNN) [91], Fast R-CNN [92], Faster R-CNN [93], and Mask R-CNN [94]. Compared with traditional computer vision algorithms, a notable advantage of these data-driven methods is their robustness to background noise and changes in external environmental conditions. The deep learning methods can automatically learn the feature representations with tens of thousands of parameters, making the feature learning process more efficient. The single-stage YOLO series, for instance, is a regression-based detector that can provide relatively high localization accuracy and precision at faster speeds. Therefore, these methods are gradually being applied to optical displacement measurements [71,95].

Cheng et al. [95] utilized the YOLO-based object detection framework to enhance the vibration measurement generalization. To accelerate computation speed, Pan et al. [97] employed the YOLOtiny and Zhu et al. [98] used the Vgg19 network to evaluate the non-contact vibration measurement capabilities. Addressing the low recognition

rates in the optical vibration measurement, Zhu et al. [99] introduced a vibration displacement measurement algorithm by enhancing the image clarity through image deblurring and feature enhancement, as shown in Fig. 6. Yang et al. [100] designed a lightweight convolutional network model using the MobileNetV2 backbone for tracking vibration displacement in rotating components. Jeong et al. [101] proposed a real-time deep-learning method to monitor structural displacement considering adverse environmental uncertainties. Huang et al. [102] and Yan et al. [103] integrated template matching with deep learning to measure the structural vibration. Furthermore, Zhang et al. [104] proposed a structural displacement monitoring method based on the Mask R-CNN to extract calibrated object coordinates.

Because deep learning methods employ artificial intelligence in vibration measurement, engineers are not required to obtain solid professional knowledge to operate the measurement systems. As a result, considerable efforts have been now made in deep learning-based optical vibration measurement.

3.3.2. Target tracking

After detecting the target points, it needs to capture the motion trajectory from the sequence of the image frames to calculate the vibration. As indicated by Huang et al. [105], the key-point information is obtained and the positions of these points are recorded to provide

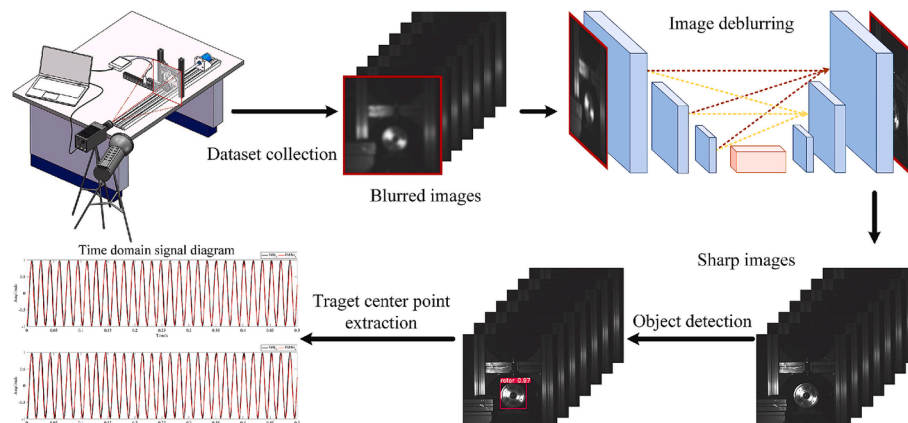


Fig. 6. The complete scheme of the applied novelty detection algorithm is based on the mean frequency maps of motion [99].

feedback on the vibration displacement. However, each frame in the sequence of images is independent, and there is no correlation between the sequence images, leading to potential frame loss or missed frames during the actual camera shooting process, which in turn causes data loss and discontinuity [96]. Therefore, establishing a correlation between images becomes particularly important.

The optical flow is a widely used technique for establishing inter-frame correlations, extensively applied in the fields of vibration measurement and motion tracking. The original methods can be divided into the differential (gradient-based) [106] and the phase-based techniques [107]. Both techniques can extract full-field optical flow information from the image sequences to calculate the image velocities and stereoscopic disparities. The differential technique relies on the assumption of constant brightness, where the projection of the same point remains consistent in the image intensity across consecutive frames. Any change indicates the motion of the point. A representative differential algorithm is the Lucas-Kanade (LK) optical flow estimation [105]. To better capture the corners or key-points, the LK method is often combined with the sparse estimation techniques. However, the LK algorithm struggles with capturing large-range motions, which led to the development of a pyramid-framework LK algorithm by Bouguet et al. [108]. Since the optical flow does not compute every single pixel, it requires relatively low computational power. For example, several studies [109–113] have used the optical flow methods combined with the feature matching techniques to track the targets and calculate the vibration. According to Zhang et al. [114], a source video is first processed through a complex filter on its first frame (reference frame) to extract the local amplitude and phase, and to determine a texture mask. Using the texture mask, integer pixel motion vectors are obtained through a pixel-matching algorithm. These motion vectors are used to generate a reconstructed video by moving the mask within the original video's first frame. Then, by comparing the original and reconstructed videos, sub-pixel motion is captured using the optical flow techniques. The total motion field of the mask pixels, which is the sum of the integer pixel motion vectors and the sub-pixel motion, ensures precise motion tracking of the video frame's textured areas, as shown in Fig. 7.

Another optical flow technique is the phase-based estimation, which is based on the assumption of the local phase constancy. This method calculates the vibration displacement using the local phase information from the images convolved with complex filters. However, this method is generally suitable only in situations with simple backgrounds. For

vibration measurement, phase-based estimation is often combined with motion magnification techniques. For instance, in [115] the edge information was extracted using the Gaussian-smoothed gradients combined with the Canny edge detection, and the phase amplification was integrated for optical flow estimation to measure the vibration of the target.

Additionally, improvements to the optical flow have been made by the RANSAC (random sample consensus) [114], MPC (mm/pixel coefficient) [117], and TDDM (tracking-detecting-deformable matching) [118] in vibration measurement. With the application of object detection in optical vibration measurement, multi-object tracking has also been developed [96,100,119], where the Kalman filtering, ByteTrack, and SORT (simple online and realtime tracking) methods are employed to correlate key information between the image frames, thereby enhancing the accuracy of the vibration measurement.

3.4. Digital image correlation (DIC)

The DIC is an optical method based on non-contact imaging, used for full-field shape, displacement, and deformation measurements [120]. This technique utilizes digital imaging devices (including but not limited to optical imaging, electronic imaging, and scanning probe imaging devices) to capture the digital images of an object under various loads (time or state). It then employs correlation-based (intensity-based or area-based) matching (tracking or registration) algorithms and numerical differentiation methods to analyze the images to quantitatively calculate the full-field displacement and strain responses [120]. According to the methodological principles, the DIC algorithms are mainly divided into the template-based regional matching methods and the speckle-pattern-based methods.

3.4.1. Dic-template region matching

Region-based template matching is a widely used optical perception technique in digital image processing. This technique involves selecting a region of interest (ROI) containing the target as a template and sliding it over another frame image to locate the moving target. During the sliding process, this algorithm searches for the highest region similarity (RS) score in subsets of the frame (i.e., overlapping areas) [121,122]. The RS score measures the similarity between the template and the area by calculating the average pixel similarity (PS) score of all pixels within the overlapping area, where the PS score assesses the similarity between

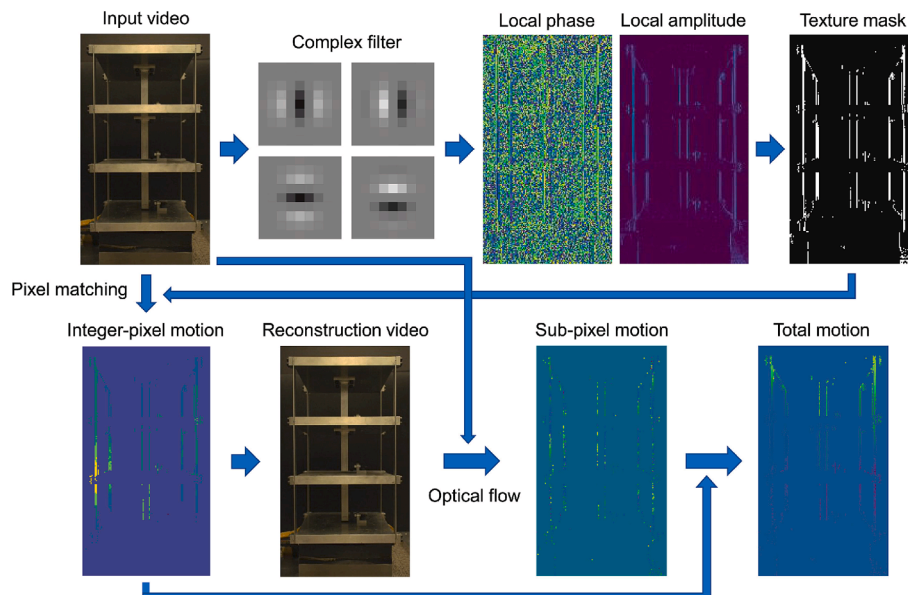


Fig. 7. The general overview processing workflow integrating optical flow technology [114].

individual pixels in the template and corresponding pixels in the overlapping area [122]. The goal of the template matching algorithm is to precisely determine the position of the template image within the global image, as shown in Fig. 8.

The template region matching, known for its simplicity, ease of implementation, and computational efficiency, has been widely used in vibration displacement monitoring. It was first applied during the 1990s for measurements at the Humber Bridge and Second Severn Crossing [124,125]. Subsequently, it has been extensively utilized for vibration displacement monitoring of railway bridges, large-span bridges, and high-rise buildings [103]. For instance, Ye et al. [126] implemented a vision-based multi-point structural displacement measurement using multi-point pattern matching. Brownjohn et al. [127] and Kimiya et al. [128] combined the optical systems with the template region matching to monitor bridge deformations. Feng and Feng [129] used the template region matching to measure the cable tension, which was preliminarily verified through laboratory tests on simply supported beams and compared with accelerometers, demonstrating good accuracy in the frequency domain. Feng et al. [64] proposed an improved template region matching that can automatically select the ROIs to estimate the structural displacement. To address potential adverse environmental effects on optical sensing systems that might cause loss of motion features in recorded video frames, Wang et al. [121] introduced a gradient-based matching and compared its performance with traditional zero-mean normalised cross-correlation and the orientation coding matching techniques to demonstrate improvement in the measurement accuracy and robustness.

3.4.2. Dic-speckle pattern

The digital speckle correlation method (DSCM) is a novel testing technique that applies digital image processing technology to photo-mechanics. The DSCM quantifies the full-field displacement and strain by comparing the grayscale values of two images of an object's surface before and after deformation [130]. The process includes several steps. First, by comparing the digital images of the surface under different states, the displacement vector of each pixel is calculated. Typically, the speckle image before deformation is referred to as the baseline or reference image, while the image after deformation is called the deformed image. Then, by finding the greatest correlation between these images, the position changes of the target (any recognizable features) from one image to another can be determined [131,132]. In digital image processing, the image deformations will affect the shape of the reference subset. According to the assumption of deformation continuity, a group of adjacent points in the reference subset will be mapped to the adjacent points in the target subset. For instance, as illustrated in Fig. 9, the centre point $P(x_i, y_i)$ of the reference subset can be mapped to the point $P'(x'_i, y'_i)$ in the target subset using a shape function or displacement mapping function.

The correlation coefficient is used to determine the similarity between the reference and target regions by comparing the grayscale values within a search region to those of the selected subset. When the correlation coefficient reaches its maximum value, it typically indicates an optimal consistency between the two regions. Therefore, choosing an appropriate correlation criterion is crucial for quantifying the similarity or dissimilarity [120].

Common correlation functions are primarily categorized into two types: based on cross-correlation and based on distance. The cross-correlation-based functions include direct cross-correlation (DCC), average cross-correlation (ACC), normalized cross-correlation (NCC), and zero-mean normalized cross-correlation (ZNCC). The distance-based functions include the Square Distance, Sum of Squared Differences, Normalized Square Distance, and Zero-mean Normalized Square Distance. Table 1 lists the four most commonly used correlation functions.

The digital speckle techniques are widely utilized across various fields such as experimental solid mechanics, civil engineering, micro-electro-mechanical systems (MEMS), and aerospace engineering [133]. During surface preparation, high-contrast speckle patterns are typically applied to structures; without these patterns, the DIC measurements of the surface deformation cannot be conducted as speckle patterns that serve as carriers of the surface deformation information.

The speckle types include natural texture speckles and artificial speckles [120]. The natural texture speckles refer to surfaces with rich natural textures and structures, making them suitable for natural feature methods. The theoretical foundation of the DIC methods used in experimental mechanics was developed in the 19th century, followed by the creation of computer programs to calculate random speckle object surface displacement components and deformation gradients caused by deformation. Experimental verification of their feasibility in experimental mechanics laid the groundwork for the engineering application of DIC technology. For instance, Pan et al. [42] applied the non-contact sensing technologies using the DIC technology on the speckles of the bridge surfaces to monitor multi-point dynamic displacements. Kim and Kim [134] proposed a DIC method for suspension cable tension measurement of the Gwang-An Bridge in South Korea. The cable dynamic responses and modal frequency resolutions can be effectively calculated. Hoult et al. [135] investigated the DIC on strain measurement to demonstrate comparable resolution to traditional strain gauges. Hieu et al. [136] developed a technology that includes the Kinect sensors, precise camera calibration schemes, and region- and feature-based image matching algorithms, to measure 3D vibration. The infrared speckle pattern projected by the Kinect sensor improved the DIC analysis efficiency.

However, the natural speckle methods suffer from low contrast, leading to potential errors in practical measurements. To enhance the target contrast and improve the accuracy of vibration measurements,

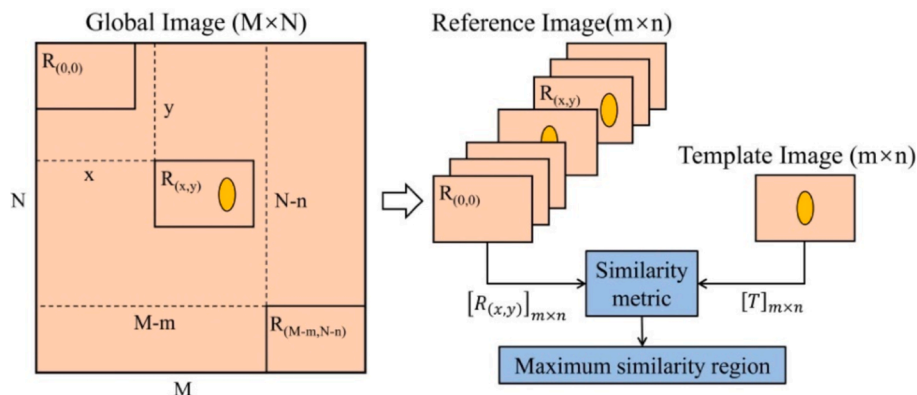


Fig. 8. Schematic of the template region matching [123].

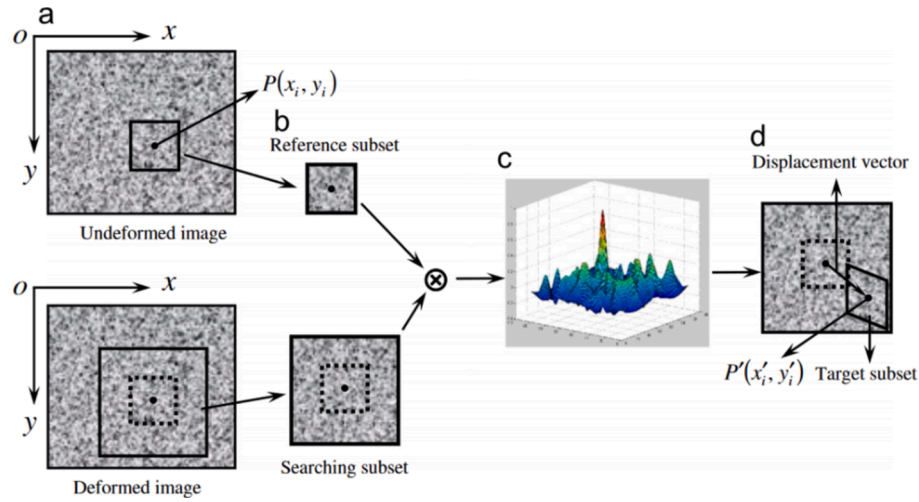


Fig. 9. Principle of digital image correlation; denotes the computation of the correlation coefficient at each location. (a) Two speckle images before and after deformation; (b) two subsets from the undeformed and deformed images; (c) distribution of the correlation coefficient; (d) displacement vector [131].

Table 1

Four types of correlation criteria used in DIC for common correlation function.

Type function	Name	Function expression
Correlation function based on cross-correlation	Direct Cross-Correlation	$C_{c1} = \sum_{i=1}^n [f(x_i, y_i) \bullet g(x'_i, y'_i)]$
	Average Cross-Correlation	$C_{c2} = \frac{1}{n} \sum_{i=1}^n [f(x_i, y_i) \bullet g(x'_i, y'_i)]$
	Normalized Cross-Correlation	$C_{c3} = \sum_{i=1}^n \frac{f(x_i, y_i) \bullet g(x'_i, y'_i)}{f_s \bullet g_s}$
	Zero-mean Normalized Cross-Correlation	$C_{c4} = \sum_{i=1}^n \frac{[f(x_i, y_i) - f_m] \bullet [g(x'_i, y'_i) - g_m]}{\Delta f \bullet \Delta g}$
Distance-based correlation function	Square Distance	$C_{d1} = \frac{1}{n} \sum_{i=1}^n [f(x_i, y_i) - g(x'_i, y'_i)]$
	Sum of Squared Differences	$C_{d2} = \sum_{i=1}^n [f(x_i, y_i) - g(x'_i, y'_i)]^2$
	Normalized Square Distance	$C_{d3} = \sum_{i=1}^n \left[\frac{f(x_i, y_i)}{f_s} - \frac{g(x'_i, y'_i)}{g_s} \right]$
	Zero-mean	$C_{d4} = \sum_{i=1}^n \left[\frac{f(x_i, y_i) - f_m}{\Delta f} - \frac{g(x'_i, y'_i) - g_m}{\Delta g} \right]$
	Normalized Square Distance	$C_{d5} = \sum_{i=1}^n \left[\frac{f(x_i, y_i) - f_m}{\Delta f} - \frac{g(x'_i, y'_i) - g_m}{\Delta g} \right]^2$

artificial speckles are introduced in industrial applications. These techniques include speckle stickers [103], sprayed speckles [137], simulated speckles [133], laser speckles [60], and adherents [138]. The speckle stickers are the most portable ones, though they may become unsuitable if cracks develop. Thus, the sprayed speckles have been developed for vibration measurement [139–141]. The simulated speckles offer the advantage of precise control over the speckle size, number, and center position, and have been widely applied in vibration measurement [133,142]. The laser speckles provide a wider depth of field and brightness control, adaptable to various sizes of measurement objects or camera resolutions, and have been extensively used for measurement at critical locations [56,60]. For larger targets as shown in Fig. 10, the high-contrast adherents are more suitable, as discussed in [142–144].

4. Summary Discussion challenges and Opportunities

The non-contact vibration measurement based on computer vision utilizes video and image analysis tools to capture and quantify the vibrational characteristics of the objects without the need for physical contact. This is particularly suitable for sensitive or difficult-to-access objects. Relying on high-speed cameras and image processing algorithms, the dynamic responses of the objects in natural environments can be monitored through continuous image sequences to extract the

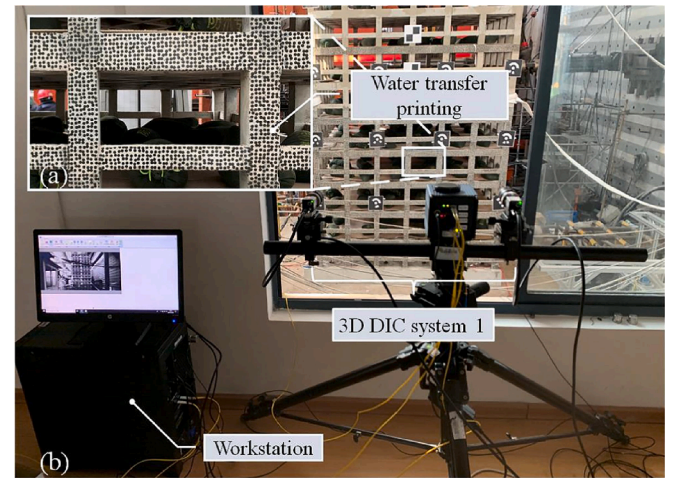


Fig. 10. (a) Speckles of water transfer printing and (b) the configuration of the 3D DIC measurement system [144].

vibration amplitude, frequency, and patterns. These applications are widespread in fields like mechanical systems, structural health monitoring, and wind turbines/blades. With the rapid progress of computer vision technology, optical vibration measurement has seen significant improvements. This review first analyzes the strengths and weaknesses of existing measurement techniques from the aspects of performance comparison, application scenarios, and vibration measurement conversion, and then provides an outlook on the challenges and future developments.

4.1. Summary-discussion

In the optical vibration measurement system, two major categories of techniques can be identified: point tracking and the DIC. Point tracking is further divided into edge point methods, gray intensity methods, and deep learning methods, as well as their hybrid. The DIC includes the template matching and speckle texture methods. Moreover, depending on the selection of target points, these methods can be classified into manual annotation methods and natural feature methods. To comprehensively evaluate the performance of each method, seven evaluation metrics have been adopted. The overall structure not only reveals the hierarchical relationships between different algorithms but also reflects

their strengths and weaknesses, thereby providing a theoretical basis for selecting appropriate algorithms. As shown in Tables 2 and 3, the classification and evaluation approach aids in the deeper understanding and application of these techniques.

Table 2 indicates that the point-tracking methods measure the vibrations by tracking specific points in the images, such as the markers or distinct features. The point tracking is straightforward and requires less computation, making it suitable for real-time monitoring and high-precision vibration measuring. When the marker points are clear and stable, the point tracking can measure local vibrations with high accuracy. However, it relies on clear, recognizable markers within the observation area, and its application is restricted to surfaces that cannot be marked or have indistinct features. Deep learning has mitigated some of these issues by reducing the need for manual parameter tuning in the feature extraction, although with the increase of computational demands. Furthermore, point tracking usually follows a few points, thus failing to provide global vibration information about a specific object.

In contrast, the DIC measures the vibration displacement and strain across the entire field of view, offering a more comprehensive vibration analysis. Because the DIC does not require any special markers on the measurement object, make it suitable for complex/sensitive surfaces.

The DIC also boasts high precision and micron-level spatial resolution, ideal for detailed analysis. However, one limitation of the DIC is the substantial computational load, which may not be suitable for real-time monitoring. Additionally, its accuracy highly depends on the image quality and stable lighting conditions, which might affect the measurement accuracy. The technical complexity of the DIC demands specialized knowledge and skills from the users.

In Table 2, a systematic statistical and detailed analysis of various measurement methods is provided to demonstrate their applications in specific fields. It can be seen that optical vibration measurement technology has become a critical tool in the field of performance testing in recent years, particularly showcasing its unique advantages in the dynamic monitoring of critical structures/equipment. Currently, the monocular vision measurement method has become the most popular choice due to its simplicity, ease of use, and accessibility. Given the distortion issues that arise with cameras in real-world applications, it is crucial to obtain both the intrinsic and extrinsic parameters of the camera for image correction, ensuring accurate measurement of the in-plane pixel vibration displacement [153]. However, the traditional approach requires fixing the distance between the camera and the structure to calculate the actual vibration displacement. In this situation,

Table 2
Algorithm Performance Comparison.

General classification	Sub-classification	Manual involvement or not		Evaluation Index						
		Manually annotated	Natural feature	1. Artificial markers: Yes or No 2. Computer performance consumption: ★ 3. Generalization: ★ 4. Feature extraction model: Human assistance or Automatic withdrawal 5. Adaptability to complex optical environments: ★ 6. Whether an association method between individual frame rates is required: Yes, No, Both 7. Measurement mode: Global measurement, local measurement, multi-point measurement.						
Point Tracking	Edge points-based	Traditional method	—	1. Yes	2. ★	3. ★	4. Human assistance	5. ★	6. Both	7. multi-point measurement
		Fusion of traditional and deep learning	—	1. Yes	2. ★★	3. ★★	4. Human assistance	5. ★★	6. Both	7. multi-point measurement
		Fully data-driven	—	1. Yes	2. ★★	3. ★★	4. Automatic withdrawal	5. ★★	6. Both	7. multi-point measurement
		—	Traditional method	1. No	2. ★	3. ★	4. Human assistance	5. ★	6. Both	7. multi-point measurement
		—	Fusion of traditional and deep learning	1. No	2. ★★	3. ★★	4. Human assistance	5. ★★	6. Both	7. multi-point measurement
		—	Fully data-driven	1. No	2. ★★	3. ★★	4. Automatic withdrawal	5. ★★	6. Both	7. multi-point measurement
	Gray Level-Based Intensity	Traditional method	—	1. Yes	2. ★★	3. ★	4. Human assistance	5. ★	6. Both	7. multi-point measurement
		Fusion of traditional and deep learning	—	1. Yes	2. ★★	3. ★★	4. Human assistance	5. ★★	6. Both	7. multi-point measurement
		Fully data-driven	—	1. Yes	2. ★★	3. ★★	4. Automatic withdrawal	5. ★★	6. Both	7. multi-point measurement
		—	Traditional method	1. No	2. ★★	3. ★	4. Human assistance	5. ★	6. Both	7. multi-point measurement
		—	Fusion of traditional and deep learning	1. Yes	2. ★★	3. ★★	4. Human assistance	5. ★★	6. Both	7. multi-point measurement
		—	Fully data-driven	1. Yes	2. ★★	3. ★★	4. Automatic withdrawal	5. ★★	6. Both	7. multi-point measurement
Digital image correlation	DIC- template area matches	—	Natural feature	1. No	2. ★★	3. ★	4. Human assistance	5. ★	6. No	7. local measurement
		Template sticker	—	1. Yes	2. ★★	3. ★	4. Human assistance	5. ★	6. No	7. local measurement
	DIC- Speckle pattern	Laser speckles	—	1. Yes	2. ★★	3. ★	4. Human assistance	5. ★	6. No	7. Global measurement
		—	Natural texture	1. No	2. ★★	3. ★	4. Human assistance	5. ★	6. No	7. Global measurement
		Sprayed speckles	—	1. Yes	2. ★★	3. ★	4. Human assistance	5. ★	6. No	7. Global measurement
		Simulated speckles	—	1. Yes	2. ★★	3. ★	4. Human assistance	5. ★	6. No	7. Global measurement
		Speckle stickers	—	1. Yes	2. ★★	3. ★	4. Human assistance	5. ★	6. No	7. Global measurement
		—	—	1. Yes	2. ★★	3. ★	4. Human assistance	5. ★	6. No	7. Global measurement

Table 3

Literature statistics table based on algorithm performance and application scenarios.

Reference	Range algorithm	Camera mode	Camera sampling frequency (Hz)	Frequency range (Hz)	Effective frequency range (Hz)	Real-time	Application scenarios
[146]	Point Tracking, Natural Feature, Fully data-driven	Monocular	2000	400	200	No	Rotating mechanical structure
[100]	Point Tracking, Natural Feature, Fully data-driven	Monocular	2000	1000	500	No	Rotating mechanical structure
[147]	Point Tracking, Natural Feature, Fully data-driven	Monocular	2000	500	250	No	Rotating mechanical structure
[99]	Point Tracking, Natural Feature, Fully data-driven	Monocular	2000	400	200	No	Rotating mechanical structure
[82]	Point Tracking, Natural Feature, Fully data-driven	Binocular	50	50	25	No	Cantilever beam
[148]	Point Tracking, Natural Feature, Fully data-driven	Binocular	100	52	25	Yes	Vibration test platform
[149]	Point Tracking, Natural Feature, Traditional method, Edge points-based	Multiple	50	50	25	No	Bridge prototype
[79]	Point Tracking, Natural Feature, Traditional method, Edge points-based	Monocular	1000	1000	500	No	Rotating mechanical structure
[70]	Point Tracking, Natural Feature, Traditional method, Edge points-based	Binocular	1000	1000	300	—	Cantilever beam
[116]	Point Tracking, Natural Feature, Fusion of traditional and deep learning	Monocular	30	30	15	No	Cabel model
[73]	Point Tracking, Natural Feature, Fusion of traditional and deep learning	Monocular	30	10	5	Yes	Cable-stayed bridge
[150]	Point Tracking, Manually annotated, Traditional method, Gray Level-Based Intensity	Monocular	29.97	29.97	14.9	Yes	Servo hydraulic motion simulator
[145]	Point Tracking, Manually annotated, Traditional method, Gray Level-Based Intensity	Monocular	100	100	50	No	Fixed steel frame
[77]	Point Tracking, Manually annotated, Traditional method, Gray Level-Based Intensity	Monocular	120	120	60	No	Vehicle-bridge model
[76]	Point Tracking, Manually annotated, Traditional method, Gray Level-Based Intensity	Monocular	120/30	120/30	60/15	No	Six-story building model
[72]	Point Tracking, Manually annotated, Traditional method, Gray Level-Based Intensity	Monocular	29.97	29.97	14.9	No	Steel model
[65]	Point Tracking, Manually annotated, Traditional method, Gray Level-Based Intensity	Monocular	25	25	—	No	Bridge
[151]	Digital image correlation, DIC- template area matches, Template sticker	Monocular	60	60	20	Yes	Bridge
[129]	Digital image correlation, DIC- template area matches, Template sticker	Monocular	25	25	12.5	No	Steel railway bridge
[64]	Digital image correlation, DIC- template area matches, Natural Feature	Monocular	150	50	25	Yes	Cable tension
[74]	Digital image correlation, DIC- template area matches, Natural Feature	Monocular	120	120	20	No	Simply supported beam
[42]	Digital image correlation, DIC- Speckle pattern, Natural texture	Monocular	—	4	2	No	Bridge
[152]	Digital image correlation, DIC- Speckle pattern, Sprayed speckles	Monocular	100	100	—	No	Cantilever beam
[144]	Digital image correlation, DIC- Speckle pattern, Sprayed speckles	Multiple	80	50	25	No	Concrete cassette structure
[56]	Digital image correlation, DIC- Speckle pattern, Laser speckles	Binocular	—	1000	1000	No	Steel plate
[133]	Digital image correlation, DIC- Speckle pattern, Simulated speckles	Binocular	80	80	—	No	Cantilever beam
[126]	Digital image correlation, DIC- Speckle pattern, Speckle stickers	Monocular	59	10	—	No	Vibration test platform

a binocular vision system may be more suitable for real-time monitoring. Furthermore, the 3D measurement techniques enhance the measurement precision through the use of binocular or multi-camera systems. A binocular vision system calculates the same scene from different directions at the same time. During this process, the intrinsic and extrinsic parameters of the two cameras and their distortion coefficients are determined through calibration. For applications that require high-precision depth information, the binocular vision measurement demonstrates its advantages. It provides more accurate depth perception capabilities without relying on fixed actual distances or the assistance of

extra displacement sensors.

For more complex and high-precision measurement requirements, such as those needing extensive coverage or high accuracy, multi-camera measurement systems are more appropriate, as illustrated in Fig. 11. These systems leverage the synchronized operation of multiple cameras, significantly enhancing the measurement's field of view and data accuracy, suitable for demanding engineering and scientific projects. As the size of the measurement objects increases, the range of traditional two-dimensional measurement methods based on single cameras and three-dimensional measurement methods based on stereo

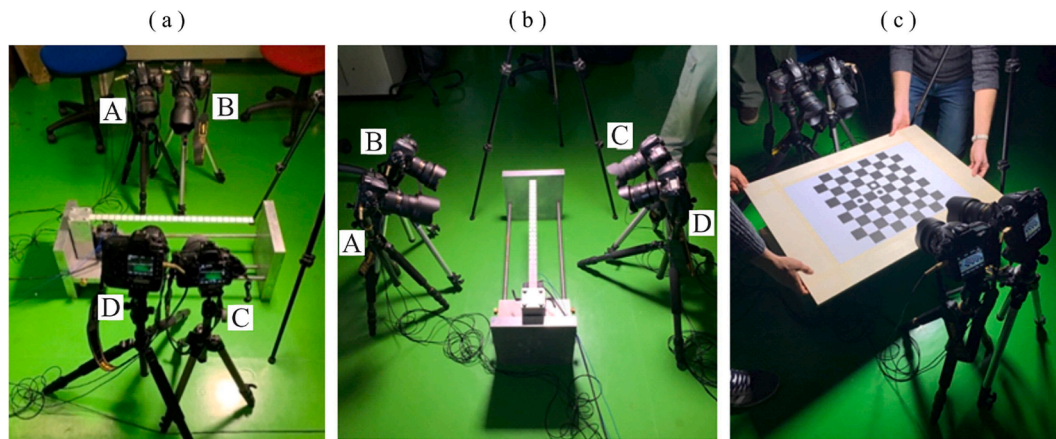


Fig. 11. Lateral (a) and front (b) view of the experimental setup. Calibration of cameras (c) [154].

cameras becomes limited. Therefore, multi-camera methods have been introduced for full-field displacement and strain measurements, which can improve accuracy while measuring large areas. For example, Xu et al. [155] developed a multi-camera DIC system based on clustering methods. Wang et al. [156] proposed a multi-camera stereo DIC device for obtaining the complete displacement and strain fields of metal plate specimens in Ericksen tests. Malesa et al. [157] employed two types of multi-camera systems to measure industrial structures. Additionally, the optical vibration measurement technology has been widely applied in the dynamic monitoring of critical structures such as cantilever beams, bridges, and rotating machinery. These applications often require the technology to not only capture the basic dynamic features of the system but also to precisely monitor minor variations to ensure system safety and operational reliability. However, current technologies face a bottleneck in high-frequency vibration measurements, mainly due to limitations in equipment processing capabilities; the highest measurement frequency typically is restricted to 128 Hz. In summary, the optical vibration measurement technology provides significant technical support for structural health monitoring, helps to detect potential structural defects and prevents catastrophic accidents. The continuous advancements in these technologies indicate an increasingly critical role in future engineering applications.

Ultimately, the choice between point tracking and DIC methods depends on the specific application requirements. If high-speed, real-time local vibration data are needed, point tracking is more suitable; however, for comprehensive, high-precision vibration analysis, especially in terms of structural stress and strain distribution, the DIC presents distinct advantages. In practical applications, these two techniques can also be combined to leverage their respective strengths and provide a more comprehensive vibration analysis performance.

4.2. Challenges and opportunities

Although various visual vibration measurement methods have been widely applied across multiple fields, many algorithms still lack universality. The fundamental reason for this lies in the differences in uncertainty across various application scenarios, as well as the challenges in standardizing and quantifying measurement accuracy, which affect the consistency and comparability of different methods. Furthermore, factors such as environmental noise, insufficient lighting, high dynamic range (HDR) heat waves, and adverse weather conditions severely interfere with image quality, reducing measurement accuracy [59]. Many scholars have addressed these issues. For example, Ye et al. [126] explored the effects of lighting, camera pitch angle, and steam on laboratory measurements, using the normalized correlation coefficient as the similarity metric for template matching. In their experiments, a fast-response QR code rigid plate was fixed on a homemade shaking table,

and five different lighting conditions (600 lx, 450 lx, 300 lx, 150 lx, and 75 lx) were set to assess the impact of lighting on the accuracy of the visual measurement system. The results showed that the measurements were highly robust to the camera pitch angle, but when the lighting intensity dropped below 75 lx, measurement accuracy was affected, and the system was sensitive to steam.

Luo [158] proposed the InnoVision algorithm, which uses a gradient-based template matching method to address the challenge of tracking low-contrast features under low-light conditions. Compared to traditional intensity-based template matching algorithms, InnoVision performs exceptionally well in recognizing and tracking low-contrast features on structural surfaces, significantly enhancing the robustness of the tracking process. Particularly in low-light environments, the algorithm demonstrates outstanding stability and strong adaptability to low-contrast features. Additionally, by introducing spline interpolation subpixel methods, the algorithm greatly improves the accuracy of displacement measurements. In the experiments, the test sites included the Manhattan Bridge and a 16.9-meter-long steel beam bridge, with the camera positioned 447 m away to capture the dynamic response of the structures. To comprehensively evaluate the performance of the InnoVision system, a vibration table was also used to simulate camera vibrations. The results showed that the vibration cancellation technique significantly reduced measurement errors, with the root mean square error (RMSE) decreasing by 61 %. Particularly in the field tests on the Manhattan Bridge, the InnoVision system demonstrated excellent multi-point displacement synchronization monitoring, fully validating its strong adaptability and precision in complex environments.

Although the above methods lay a solid research foundation in terms of environmental interference, they still fall short in the unification of vibration measurement. Algorithmic unification not only helps enhance the overall understanding of specific domains but also promotes the advancement of academic research, accelerating the breaking down of existing technological barriers. In recent years, deep learning, with its exceptional generalization capability [159], has significantly improved vibration detection performance in complex lighting conditions and HDR scenes. Deep learning has become a key force in achieving algorithmic unification and is expected to play an important role in the following five technological breakthroughs.

- (1) **High generalization capability of labels:** With the introduction of the concept of datasets, deep learning has provided a new direction to overcome the limitations of traditional visual vibration measurement methods and is expected to further enhance the versatility of visual vibration measurement systems. Compared to natural feature-based methods, vision-based vibration measurement using manually labeled markers is more stable, making the development of high-contrast labels particularly important. In

visual vibration measurement, labeling methods are primarily divided into passive optical markers and active optical markers. As shown in Section 3.1, passive optical markers include QR codes, circular targets, checkerboards, etc., which rely on external light source reflection or processing by imaging systems for recognition. On the other hand, active optical markers include LED light sources, laser point sources, and laser spots, which provide visual information through active emission and are typically used in applications that require high precision and stability.

Currently, the existing labeling systems are still fragmented and lack standardization. By employing a unified mathematical image processing approach, we can evaluate existing labels and identify the most advantageous ones to aid in the standardization process. Additionally, it is important to design various types of unified labels tailored to different vibration measurement targets. To enhance the generalization capability of labels, it is crucial to standardize label types and create corresponding datasets based on specific application requirements. This will enable future researchers to perform more effective algorithm evaluations.

- (2) **Efficient target detection algorithms:** With the development of deep learning, the generalization ability of object detection technology has significantly improved, providing greater environmental adaptability for mechanical structure vibration measurement. It also enables rapid localization of local regions, thereby offering more precise local information for subsequent algorithms and improving measurement accuracy. As a result, these methods have been widely applied in vibration measurement, as demonstrated by the fully data-driven (Deep Learning) approach mentioned in Section 3.3.1. However, despite the significant progress these methods have made in enhancing structural vibration measurement accuracy, inference speed remains relatively low in practical applications, posing a significant limitation for real-time vibration measurement. Additionally, existing research has relatively few contributions to algorithm generalization, and it is difficult to quantify their adaptability. Therefore, future research needs to focus on developing algorithms that are both high-precision and high-speed to meet the demands of embedded devices. At the same time, adopting unified datasets and standardized labeling systems will help improve the generalization of these methods, promoting their application in a broader range of fields.
- (3) **Effective target tracking methods:** Keypoint tracking methods can accurately extract vibration information by precisely capturing and analyzing the displacement of feature points in images. They perform exceptionally well, especially in scenarios involving small vibration amplitudes or high-speed vibrations. In Section 3.3.2, we reviewed the application of traditional optical flow methods in vibration measurement. However, traditional optical flow methods typically rely on certain physical assumptions, such as the brightness constancy assumption, and estimate optical flow by calculating pixel displacements in images. These methods are mainly based on image gradients and local features. In contrast, deep learning-based optical flow methods use neural networks to automatically learn image features and motion patterns, allowing them to more effectively handle complex scenes and dynamic changes, such as variations in lighting, occlusions, and texture loss [160,161].

In this field, FlowNet [162] represents a pioneering deep-learning model for optical flow. Building on this, Ilg et al. improved FlowNet by stacking multiple modules, resulting in FlowNet 2.0 [163], which was the first deep learning model to surpass traditional algorithms in terms of performance. However, despite the breakthrough in accuracy,

inference speed remained relatively slow. Subsequently, Sun et al. further integrated classical techniques, such as pyramid processing, deformation, and cost volume calculation, into the model, proposing PWC-Net [164]. This improvement significantly enhanced the efficiency of inference speed. Similarly, Hui et al. introduced LiteFlowNet [165] by combining pyramid processing and optical flow regularization, which led to a substantial increase in inference speed.

Although the above methods have made significant progress in reducing model size and improving inference speed, they still struggle to fully meet practical demands, as visual vibration measurement typically involves the combination of multiple algorithms. Therefore, designing keypoint tracking methods that are both fast and efficient, based on a universal visual vibration measurement framework, remains a significant challenge.

- (4) **Auxiliary accuracy enhancement methods:** Mechanical vibration is typically manifested as small, oscillatory displacements, particularly in high-frequency vibrations where the amplitude is often very small. However, most traditional computer vision (CV) techniques are primarily designed for larger amplitude vibrations, making them less suitable for the precise measurement of small-amplitude vibrations [37]. Motion magnification methods amplify the subtle motion signals in videos, making tiny vibrations that are difficult to detect with the naked eye visible. This technique has been widely applied in fields such as offshore wind turbines [166], bridge cables [167,168], and bridges [169]. On the other hand, subpixel methods enhance the accuracy of pixel positioning in images or videos, offering higher localization precision than traditional pixel-level measurements. This is particularly important for vibration measurement in low-resolution images or videos and has been used in applications such as bridges [73,170] and cantilever beams [171].

However, motion magnification methods also have some limitations, such as being sensitive to noise, having high computational overhead, potentially uneven amplification effects, and being prone to confusion with other motion features. At the same time, subpixel methods face challenges such as high computational complexity, sensitivity to noise, and dependence on image quality. In practical applications, how to choose the appropriate technique based on vibration characteristics, image quality, and computational resource constraints remains key to improving the accuracy of low-amplitude vibration measurement. Additionally, the field of view is typically limited by the number of pixels. Although expanding the field of view can increase the coverage area, it also increases computational load. Conversely, reducing the field of view can improve measurement speed but limits the coverage area. Therefore, how to achieve high-precision vibration measurement within a limited pixel range remains an important issue that needs to be addressed.

- (5) **Multi-camera switching:** In the process of visual vibration information processing, pixel displacement is first calculated, and then the pixel displacement is converted into actual vibration displacement. This process requires camera calibration and the solving of three-dimensional spatial displacement. Traditional monocular vision suffers from lower accuracy due to the lack of depth information and is more sensitive to changes in lighting [154]. While binocular vision can provide depth information, it is more computationally complex, has higher scene requirements, and comes with a higher system cost. Multi-camera vision systems are even more complex, with very high requirements for scene texture and calibration accuracy, as well as greater demands on computational resources. These limitations restrict the application of traditional visual technologies in high-precision and complex scenes.

In recent years, significant progress has been made in monocular depth estimation using deep learning methods. By leveraging convolutional neural networks (CNNs) and large datasets of images paired with corresponding depth maps, depth prediction can be performed effectively [172]. This approach has relatively low computational complexity, minimal hardware requirements, and is suitable for low-cost, portable devices. For example, Shao combined a monocular vision depth estimation neural network with a novel 3D data generation technique and advanced large-scale visual segmentation models, successfully applying it to full-field out-of-plane displacement measurements of civil structures [173,174], with the underlying network architecture shown in Fig. 12. However, the performance of this method may be limited in complex or texture-poor scenes. In contrast, binocular depth estimation methods, when combined with deep learning, automatically learn the disparity information between image pairs through deep neural networks, providing higher accuracy and direct absolute depth information. However, this method typically requires more computational resources and relies on a dual-camera system [175]. Multi-view depth estimation builds on the principles of binocular depth estimation, using multiple cameras to capture images from different viewpoints and establish correlations between the views to further improve depth estimation accuracy [176]. However, it has the highest computational complexity and hardware demands among all the methods.

Monocular depth estimation holds great potential in visual vibration measurement but still requires substantial computational resources when applied to high-frequency vibration measurements. With the improvement in computing power, depth estimation is expected to become an important trend in this field. By introducing monocular depth estimation, pixel displacement can be converted into 3D vibration displacement, providing a foundation for subsequent optimization and improving measurement accuracy. The system can also dynamically adjust based on the number of cameras, with monocular algorithms sufficient for simple scenarios and multi-view fusion improving accuracy in more complex ones. This design enhances system flexibility and performance.

5. Conclusions

Optical vibration measurement systems, as a tool for engineering vibration measurement, exhibit significant potential for application in the engineering field thanks to their cost-effectiveness, convenient installation process, and superior performance. These systems not only offer a broad frequency measurement range but also possess excellent spatial resolution, making them an ideal choice for assessing the condition of aging engineering structures and infrastructure. This review provides a comprehensive review of the latest research developments in optical vibration measurement methods and their widespread

applications. The main conclusions drawn are as follows.

- (1) Based on the principles, they are mainly divided into point-tracking methods and digital image correlation methods. Both are applicable to different structures and can provide effective solutions when markers or natural features are present.
- (2) Whether it is traditional methods, data-driven methods, or a combination of traditional and data-driven methods, the use of manual labels can improve the model's generalization ability and enhance its resistance to environmental interference.
- (3) To achieve generalized vibration measurement based on computer vision, it is particularly important to construct a labeled dataset with high generalization ability, or a dataset that covers multiple categories. This not only helps unify the application of algorithms but also relatively reduces the complexity of deep learning algorithms, improving inference efficiency.
- (4) Despite the rapid development of the computer industry, the efficiency of the algorithm itself remains a significant challenge in applications involving fast multi-point measurements. How to improve the processing speed and real-time performance of the algorithm while ensuring measurement accuracy is still a key issue in current technological development.
- (5) Due to the strong generalization ability of deep learning methods, they can maintain good performance in various complex scenarios, providing stronger support for handling diversified tasks and managing complex environmental interference. Therefore, they will still play an important role in future generalized measurement applications.
- (6) Future vibration measurement algorithms will be based on unified fundamental algorithms, capable of flexibly integrating various camera information according to different application scenarios and requirements to improve estimation accuracy. These systems will have high scalability, able to dynamically adjust based on hardware resources and computational needs, ensuring optimal performance in various environments. This will become an important development trend.

CRedit authorship contribution statement

Yuansheng Cheng: Writing – original draft, Methodology, Funding acquisition, Formal analysis. **Zhe Tian:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Donghong Ning:** Writing – original draft, Project administration, Methodology, Investigation. **Ke Feng:** Writing – original draft, Investigation, Data curation, Conceptualization. **Zhixiong Li:** Writing – review & editing, Visualization, Supervision. **Sumika Chauhan:** Writing – review & editing. **Govind Vashishtha:** Writing – review & editing, Methodology.

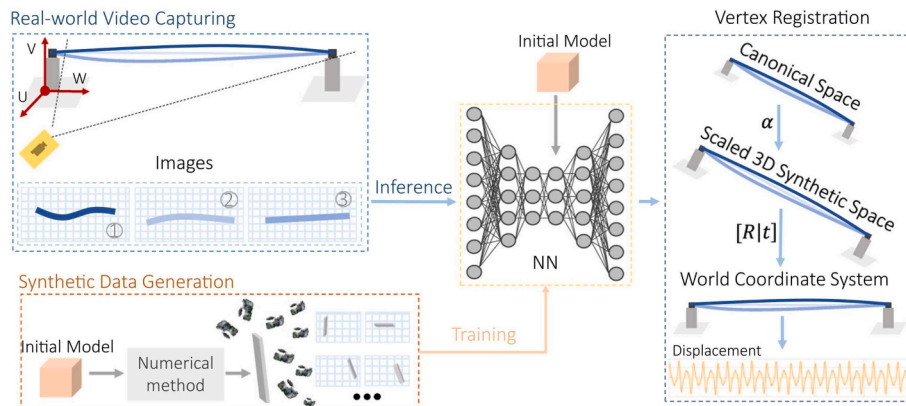


Fig. 12. The overview of the proposed displacement measurement system. [172].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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